jupyter Untitled2 Last Checkpoint: 7 minutes ago

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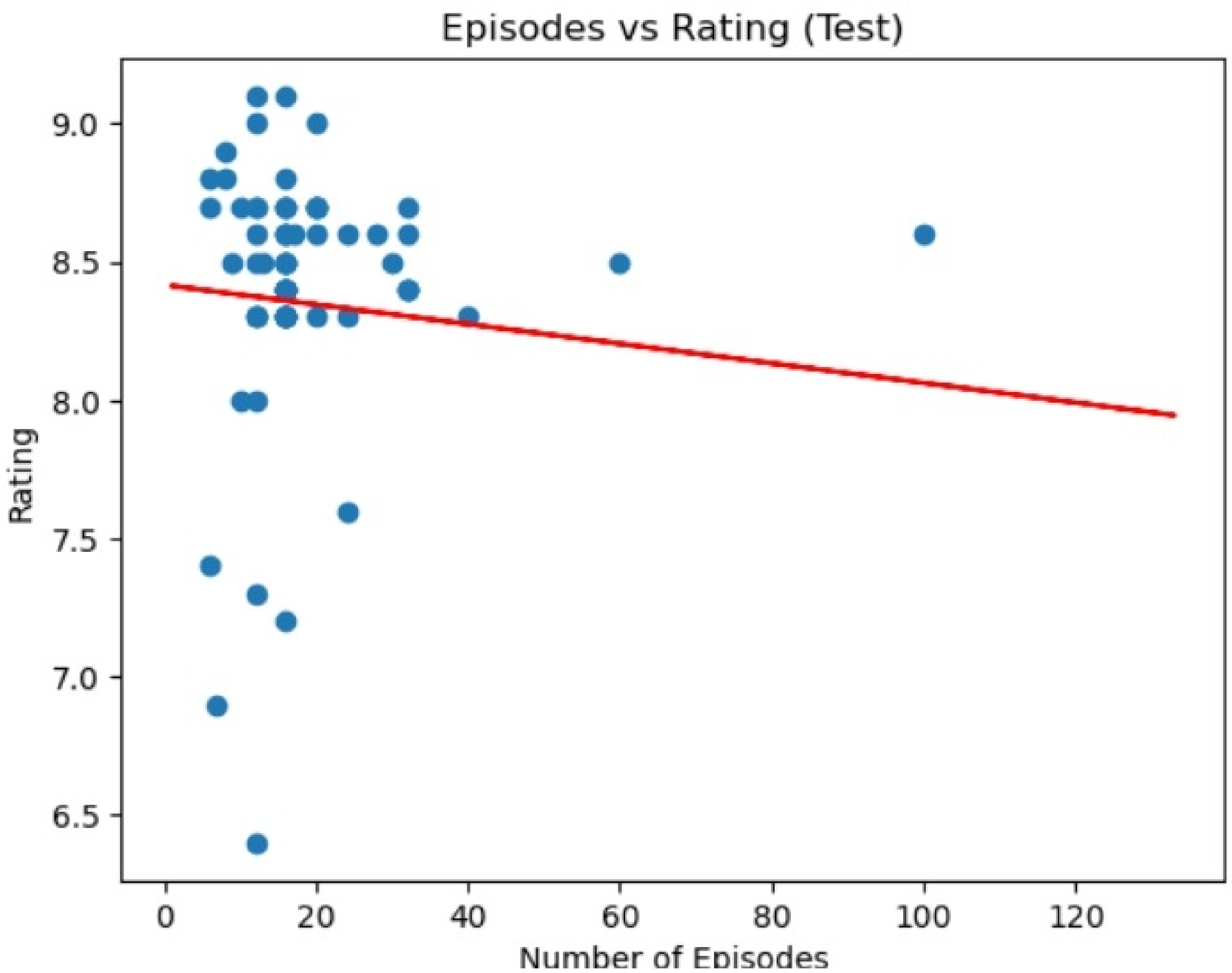
Code

JupyterLab Python 3 (ipykernel)

[1]:

```python
# Import libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

# Load dataset
dataset = pd.read_csv("kdrama_DATASET.csv")

# Select features
X = dataset.iloc[:, 3:4].values      # Number of Episodes
y = dataset.iloc[:, 4].values        # Rating

# Split dataset
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=0
)

# Train Simple Linear Regression
from sklearn.linear_model import LinearRegression

lr = LinearRegression()
lr.fit(X_train, y_train)

# Predictions
y_pred = lr.predict(X_test)
```

[4]:
```
plt.scatter(X_test, y_test)
plt.plot(X_train, lr.predict(X_train), color='red')
plt.title("Episodes vs Rating (Test)")
plt.xlabel("Number of Episodes")
plt.ylabel("Rating")
plt.show()
```



[3]:

```
plt.scatter(X_train, y_train)
plt.plot(X_train, lr.predict(X_train), color='red')
plt.title("Episodes vs Rating (Training)")
plt.xlabel("Number of Episodes")
plt.ylabel("Rating")
plt.show()
```

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Code JupyterLab Python 3 (ipykernel)

```
   Actual  Predicted
0     9.1   8.373961
1     8.8   8.359778
2     8.3   8.373961
3     8.8   8.388143
4     8.0   8.373961
```

[2]:

```
from sklearn.metrics import r2_score, mean_absolute_error, mean_squared_error
import numpy as np

print("R2 Score:", r2_score(y_test, y_pred))
print("MAE:", mean_absolute_error(y_test, y_pred))
print("RMSE:", np.sqrt(mean_squared_error(y_test, y_pred)))
```

```
R2 Score: -0.05129827778228324
MAE: 0.31959659840521965
RMSE: 0.47420292133669045
```